PIYUSH CHAUHAN

Junior Software Engineer | React.js · TypeScript · Python · FastAPI · WebSockets · PostgreSQL · AI Integrations

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PROFESSIONAL SUMMARY

Results-driven Junior Software Engineer with practical industry exposure building modern web applications, asynchronous backend services, and scalable full-stack architectures. Currently interning at CodeGarage Tech, actively contributing to production engineering including an AI Codebase Architecture & Vulnerability Visualizer (React, TypeScript, React Flow, WebSockets, Python, FastAPI, ELK) and customer communication workflows integrated with Chatwoot. Proven capability engineering end-to-end web applications with modern AI tooling (Antigravity, Cursor, Claude Code, Gemini AI), real-time bi-directional streaming via WebSockets, and interactive graph systems. Proficient in scalable database design (PostgreSQL, MongoDB, MySQL), RESTful API engineering, modern state management, and systematic debugging across complex codebases.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frontend & UI: React.js, React Flow, Next.js, Responsive Web Design, Component Architecture, Material UI

Backend & APIs: FastAPI, Python Async (async/await), WebSockets, RESTful APIs, Microservices, Node.js Basics

AI Tools: Antigravity, Claude Code, Cursor, LLM Prompting & API Orchestration, Gemini AI

Databases & ORMs: PostgreSQL, MongoDB, MySQL, SQLAlchemy, SQLite, Sequelize

DevOps, Tools & OS: Git, GitHub, VS Code, ELK Stack (Elasticsearch, Logstash, Kibana), Postman, Linux (Ubuntu), Windows

Core CS Concepts: OOP, Data Structures & Algorithms, DBMS, CRUD Operations, Debugging & Troubleshooting

Web & Mobile (Secondary): Cross-Platform UI Basics, Flutter & Dart

CORE COMPETENCIES

- Full-Stack Web & API Architecture
- AI-Assisted Dev (Antigravity & Cursor)
- RESTful API Design & Webhooks
- Systematic Debugging & Profiling
- Interactive Graph & Flow UI (React Flow)
- AI & LLM Integrations (Claude / Gemini)
- Authentication & Security (OTP, JWT)
- Observability & Log Tracing (ELK)
- Real-Time WebSockets & Streaming
- Relational & NoSQL Database Design
- Asynchronous Microservices (FastAPI)
- Git Version Control & Agile Delivery

PROFESSIONAL EXPERIENCE

Software Engineering Intern — CodeGarage Tech

Mohali, Punjab | March 2026 – Present

TypeScript · React.js · React Flow · Python · FastAPI · WebSockets · ELK Stack · PostgreSQL · MongoDB · Chatwoot

Actively contributing to production engineering across two live enterprise client platforms, applying scalable software architecture, structured modular folder conventions, and modern full-stack software engineering practices.

Live Project 1: AI Codebase Architecture & Vulnerability Visualizer

- Collaborate on an automated developer tool that ingests user-uploaded repositories or connects via GitHub, scanning AST syntax trees to generate interactive architectural dependency diagrams and highlight structural bugs and security weaknesses.
- Implement interactive graphical node-and-edge flow components using React, TypeScript, and React Flow, enabling intuitive code navigation for complex software systems.
- Integrate WebSockets for real-time bi-directional progress streaming, notifying the frontend during long-running repository scanning and background worker execution.
- Work on backend microservices powered by Python and FastAPI; assist in logging, tracing, and runtime observability using the ELK (Elasticsearch, Logstash, Kibana) stack.
- Conduct in-depth debugging, code profiling, and modular refactoring across complex multi-file frontend and backend repositories.

Live Project 2: Omnichannel Customer Calling & Billing Platform (Chatwoot)

- Manage and maintain the customer billing and subscription management module for an enterprise customer communication and voice calling application built on Chatwoot.
- Monitor billing transaction logs, audit customer call metering records, verify payment status synchronizations, and resolve accounting logic anomalies.
- Utilize strict Git/GitHub branching workflows, participate in sprint standups, and coordinate bug resolution with senior developers.

Software Engineer Intern — Novem Control Private Limited

Yamunanagar, Haryana (Hybrid) | June 2025 – August 2025

Flutter · Dart · Sqflite · Hive · Firebase · Material Design · Git

- Engineered responsive cross-platform mobile application modules using Flutter and Dart with strict focus on UI performance, widget lifecycle management, and Material Design guidelines.
- Implemented high-speed local data persistence and offline-first storage mechanisms using Sqflite and Hive, ensuring zero data loss during network disruptions.
- Integrated Firebase Firestore and Authentication for real-time backend synchronization, user session security, and dynamic cloud data handling.
- Collaborated directly with senior mentors to isolate runtime exceptions, optimize UI widget build times, and maintain clean separation of concerns using MVC/MVVM patterns.

FEATURED TECHNICAL PROJECTS

AI-Powered Full-Stack Blog Platform with Secure OTP Authentication *Personal Project* | ***cortex-blogs.piyushchauhandev.in ↗***

React.js · TypeScript · Python · FastAPI · PostgreSQL · SQLAlchemy · Google Gemini AI · JWT · OTP Service

- Architected and deployed an end-to-end full-stack publishing platform featuring a responsive modern UI, markdown article authoring, and robust user role access controls.
- Integrated Google Gemini AI API to provide automated article summarization, intelligent content tagging, and AI-assisted drafting suggestions for writers.
- Engineered a secure email-based authentication system featuring time-sensitive One-Time Passwords (OTP), bcrypt hashing, and rotating JSON Web Tokens (JWT) for session security.
- Built high-throughput asynchronous RESTful APIs using FastAPI with strict Pydantic schemas and PostgreSQL relational storage managed through SQLAlchemy ORM.
- Optimized database queries with indexing, implemented CORS security middleware, and deployed the full application to live cloud hosting (**cortex-blogs.piyushchauhandev.in**).

EasyShare – Real-Time Peer-to-Peer File & Data Sharing Application *Mobile Engineering Project*

Flutter · Dart · WebRTC · Firebase Firestore · Asynchronous Streams · Material 3

- Engineered a high-performance cross-platform application for seamless device-to-device file transfers and real-time data exchange using WebRTC peer-to-peer protocols.
- Achieved ultra-low latency direct data streaming between mobile devices, eliminating server bandwidth bottlenecks and ensuring end-to-end user privacy.
- Utilized Firebase Firestore for dynamic signaling, session negotiation (SDP offer/answer), ICE candidate exchange, and room management.
- Implemented secure room pairing through auto-generated unique Room IDs, dynamic connection state monitors, and automated reconnection handlers.
- Designed a streamlined Material 3 user interface with live transfer progress indicators, chunked data handling, and file integrity validation.

EDUCATION

Master of Computer Applications (MCA) 2024 – 2026
Tilak Raj Chadha Institute of Management and Technology (TIMT), Yamunanagar **CGPA: 8.4 / 10.0**

Post Graduate Diploma in Computer Applications (PGDCA) 2022 – 2024
Government College, Chhachhrauli **Score: 67.1%**

Bachelor of Science (B.Sc. – Non-Medical) 2018 – 2022
Mukand Lal National College, Yamunanagar **Score: 69.2%**

Senior Secondary Certificate (Class XII – Non-Medical) 2018
Haryana Board of School Education (HBSE) **Score: 62.2%**

Secondary School Certificate (Class X) 2016
Haryana Board of School Education (HBSE) **Score: 75.2%**

CERTIFICATIONS & ACCREDITATIONS

Full-Stack Web Development & APIs: Complete Coding
Core Concepts, OOP Principles, REST APIs & Algorithms

MySQL Database Specialization: Coursera
Relational Database Design, Indexing & Query Tuning

Modern JavaScript Development: Let's Upgrade
ES6+, Asynchronous Programming, Promises & DOM

Git & GitHub Version Control: Let's Upgrade
Branching Strategies, Merge Conflicts & Collaboration

CCDL (Certificate in Desktop Literacy): HARTRON Jagadhri
Information Technology & Computing Essentials

NSQF Level 1, 2, 3, 4 Certification: NSDC
National Skill Development Corporation Accredited